

Metropolis Algorithm for Yang-Mills Matrix Models

Project Report

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1 Yang-Mills Matrix Model

Yang-Mills theory is a gauge theory based on special unitary group. Yang-Mills theory seeks to describe the behavior of elementary particles. It is at the core of the theory of strong nuclear force. It also appears in the theory of weak interactions. Thus it forms the basis of our understanding of the Standard Model of particle physics. Here we will only deal with the zero-dimensional version of the model.

1.1 The Yang-Mills action in zero dimensions

The action is as follows

$$S_{YM} = -\frac{N}{4\lambda} \text{Tr} \sum_{\mu, \nu}^d [X_\mu, X_\nu]^2. \quad (1)$$

Here S_{YM} is the action, X 's are $N \times N$ Hermitian matrices, $\mu, \nu = 1, 2, \dots, d$, and $\lambda = g^2 N$.

For $d = 4$ case we can get the action in the following way. In $3+1$ dimensions we have

$$S = \int d^4x \mathcal{L} = -\frac{1}{4g_4^2} \int d^4x \text{Tr} \sum_{\mu, \nu}^d F_{\mu, \nu} F_{\mu, \nu}, \quad (2)$$

where g_4 is the four dimensional coupling and it has E^0 dimensions.

Dimensionally reducing the Lagrangian to zero spacetime dimensions we get

$$\mathcal{L} = -\frac{1}{4g^2} \text{Tr} \sum_{\mu, \nu}^d [X_\mu, X_\nu]^2. \quad (3)$$

The zero-dimensional coupling g has E^2 dimensions. The action S in the same as the Lagrangian since there is no spacetime to integrate over. Thus

$$S = -\frac{1}{4g^2} \text{Tr} \sum_{\mu, \nu}^d [X_\mu, X_\nu]^2 \quad (4)$$

$$= -\frac{1}{4\lambda} \text{Tr} \sum_{\mu, \nu}^d [X_\mu, X_\nu]^2. \quad (5)$$

Rescaling the fields $\frac{X}{\sqrt{\lambda}} \rightarrow X$, we get

$$S = -\frac{N}{4} \text{Tr} \sum_{\mu, \nu}^d [X_\mu, X_\nu]^2. \quad (6)$$

1.2 Analytical Result

In analytical result is

$$\frac{\langle S \rangle}{N^2 - 1} = \frac{d}{4}. \quad (7)$$

Here S_{YM} is the action, N is the number of colors and d is the dimension of spacetime before compactification.

We verify this exact result with the numerical result generated using Metropolis algorithm.

2 Markov Chain Monte Carlo

2.1 Markov Chains

A Markov chain is a sequence of events which a certain property. The property states that probability of next state depends only on the present state and none of the ones that came previously. To make it more precise, the us consider the following notations. Let $S = s_1, s_2, s_3, \dots, s_i, \dots, s_n$ be the countable set of states of the system. S is called the *state space*.

We have

$$\begin{aligned} P[X_{t+1} = s_i | X_t = s_{i-1}, X_{t-1} = s_{i-2}, X_{t-2} = s_{i-3}, \dots] \\ = P[X_{t+1} = s_i | X_t = s_{i-1}]. \end{aligned}$$

Here we define X_t to be state of the system at time t , and if it follows this given condition then state is said to satisfy the Markov property.

This statement just shows that future evolution is dependent only on the present state.

Here the P_{ij} , denotes the probability the system will change its state from s_i to s_j and is called the *transition probability*.

2.2 Monte Carlo Method

Monte Carlo simulations are named after the popular gambling destination in Monaco, as the method relies on use of randomness, much like it is present in gambling.

The technique was first developed by Stanislaw Ulam, who was a Polish-American scientist in the fields of mathematics and nuclear physics.

The basis of a Monte Carlo simulation is that the probability of varying outcomes cannot be determined because of random variable interference. Therefore, a Monte Carlo simulation focuses on constantly repeating random samples to achieve certain results. An uncertain variable is taken and assigned a random value. The model is then run and a result is provided. The simulation runs for a number of iterations and the average is provided as the final result.

2.3 Metropolis Algorithm

1. First to calculate the action, we need the field configurations X . Initial field configuration can be chosen in two ways. A random start is known as a *hot start*. If initial field is set to zero then it is called a *cold start*.
2. Next, make a random change in the field and calculate the new action.
3. Calculate the change in action $\Delta S = S_{\text{new}} - S_{\text{old}}$
4. Next we do the accept/reject process. Is $\Delta S \leq 0$, the field configuration is accepted.
5. If $\Delta S > 0$, then choose a random number u in the interval $[0, 1]$ and if $r < \exp -\Delta S$ then accept the field configuration, otherwise reject it.
6. If a configuration is accepted then the accepted ones become the old fields.
7. This process is then repeated multiple times and the accuracy of our result depends on how many times we repeat it.
8. Out of the total iterations performed, we set some for thermalization, and using the rest of the values we calculate the average of the statistical observable.

3 Simulation Results

3.1 Acceptance Rate

We use the Metropolis algorithm to calculate the observable $\langle S \rangle$, that is the average of the action. The generated data is compared with the analytical solution. The code is written in Python and the step size is set such that the acceptance rate is in between 60% - 90%.

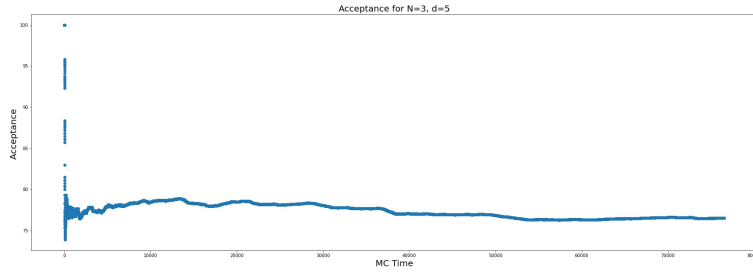


Figure 1: Acceptance rate for $N = 3$ and $d = 5$.

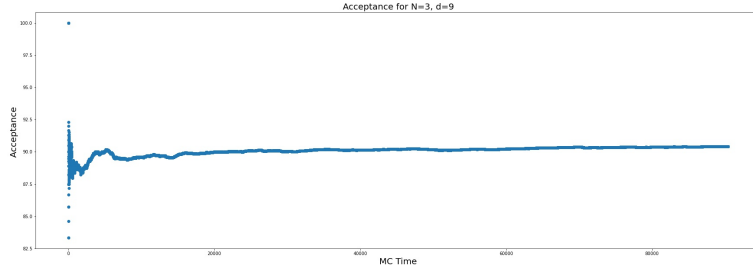


Figure 2: Acceptance rate for $N = 3$ and $d = 9$.

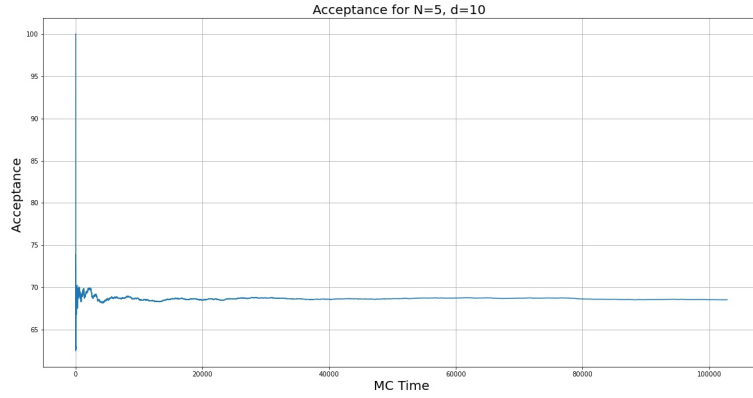


Figure 3: Acceptance rate for $N = 5$ and $d = 10$.

3.2 Observable (Action)

The simulations were done for various values of N and d and the action was computed and the results are shown below.

3.2.1 Thermalization

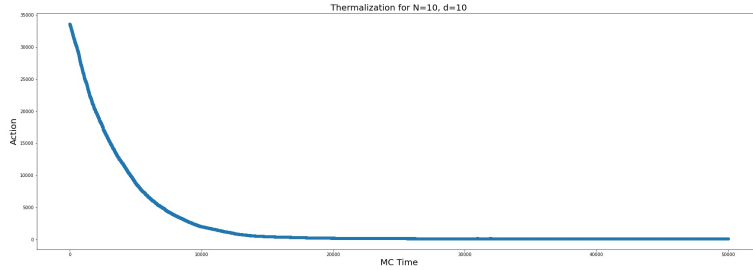


Figure 9: Thermalization for $N = 10$ and $d = 10$.

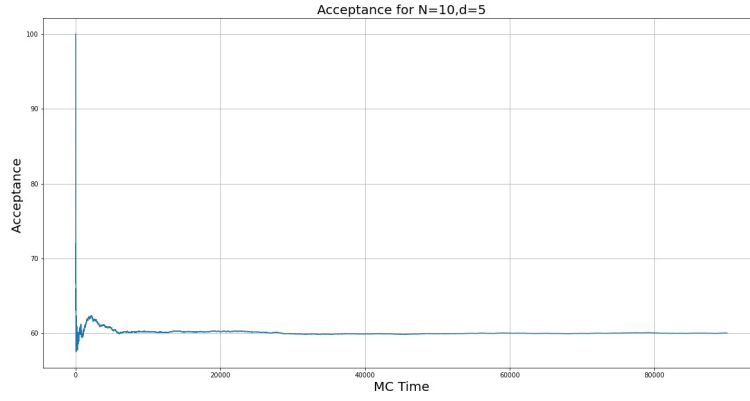


Figure 4: Acceptance rate for $N = 10$ and $d = 5$.

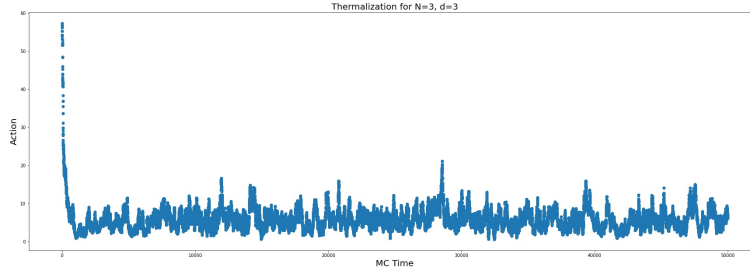


Figure 5: Thermalization for $N = 3$ and $d = 3$.

Here we can see that both cold and hot start give us properly thermalized values. At times a hot start maybe be a very large value and due to scale of the graph it looks like its approaching zero, but the thermalized action values in the next section show that these are indeed non-zero.

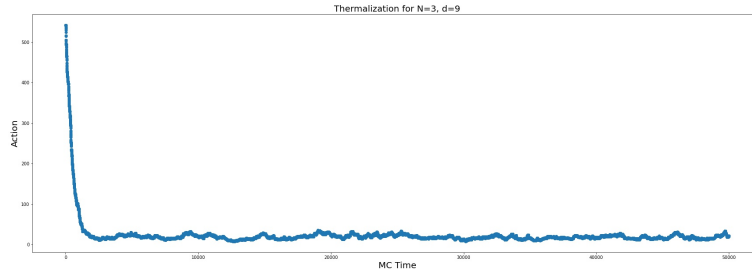


Figure 6: Thermalization for $N = 3$ and $d = 9$.

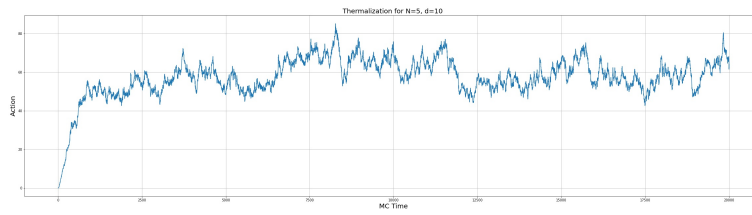


Figure 7: Thermalization for $N = 5$ and $d = 10$.

3.3 Thermalized Values

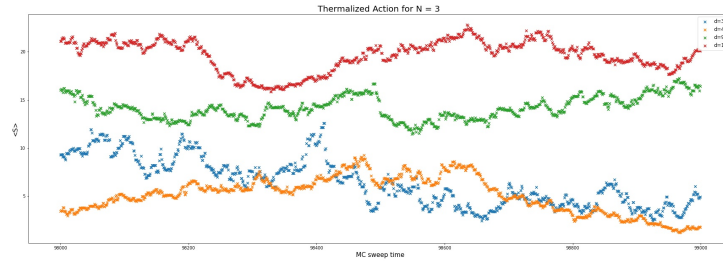


Figure 10: Thermalized action for $N = 3$.

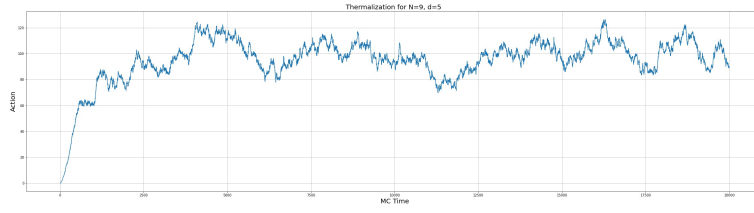


Figure 8: Thermalization for $N = 9$ and $d = 5$.

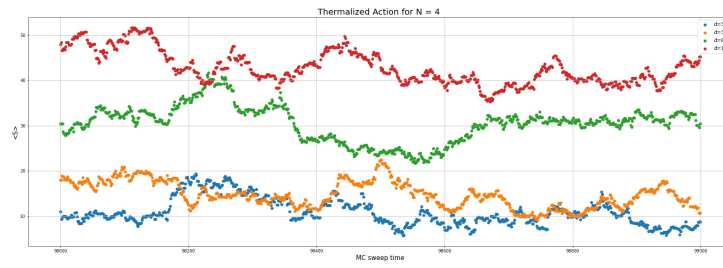


Figure 11: Thermalized action $N = 4$.

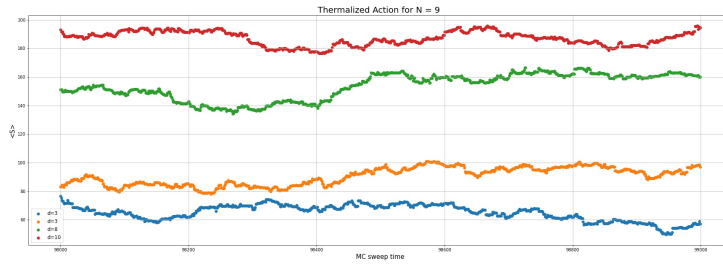


Figure 12: Thermalized action $N = 9$.

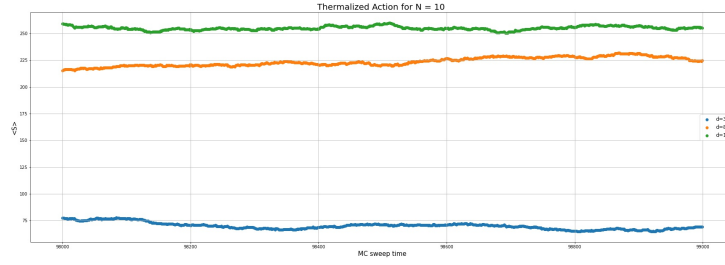


Figure 13: Thermalization for $N = 10$.

Here Fig. 10 to Fig. 13 show the values of action after the thermalization process.

3.3.1 Average Action vs d

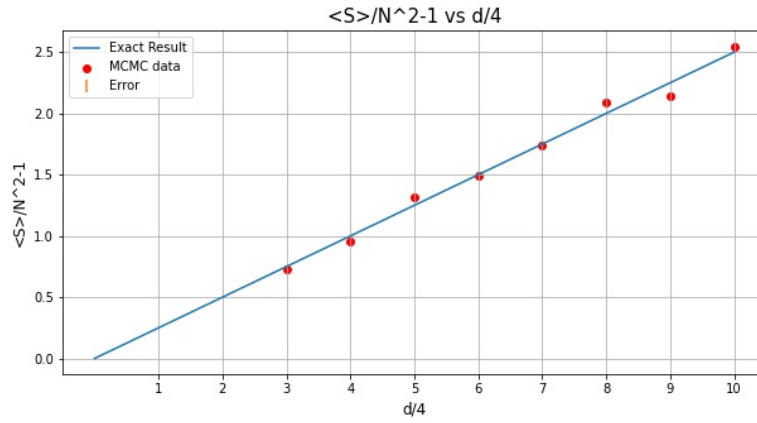


Figure 14: Plot for $N = 3$.

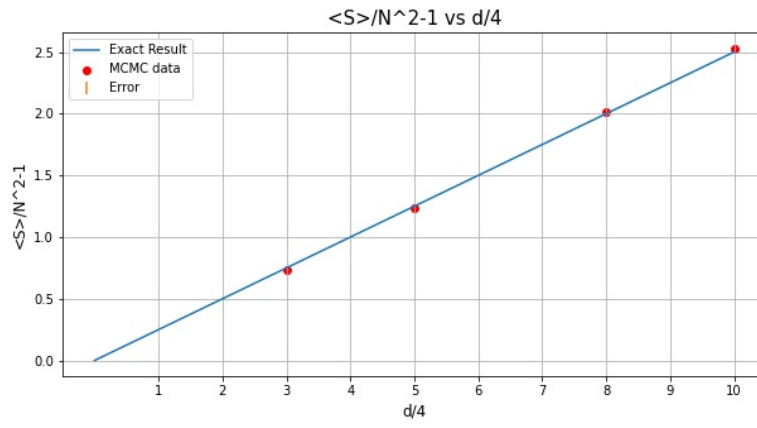


Figure 15: Plot for $N = 4$.

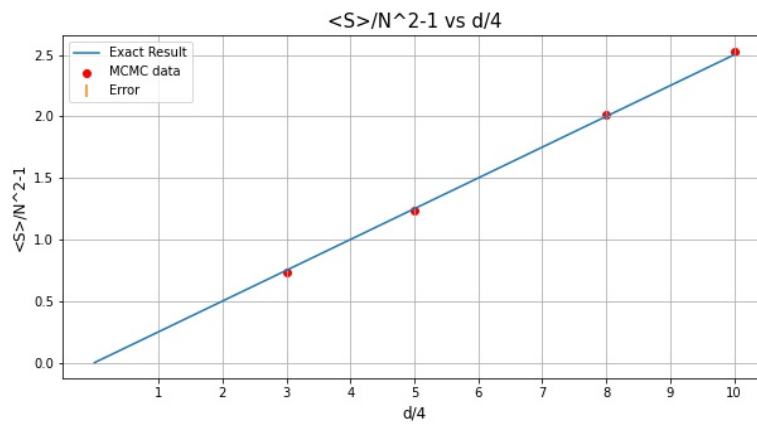


Figure 17: Plot for $N = 9$.

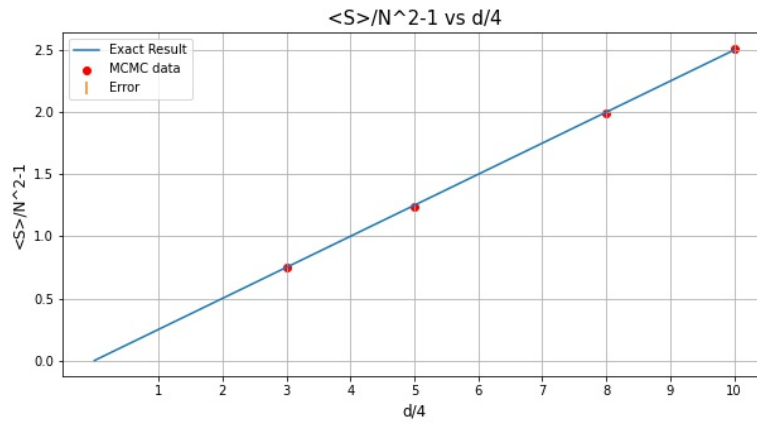


Figure 16: Plot for $N = 5$.

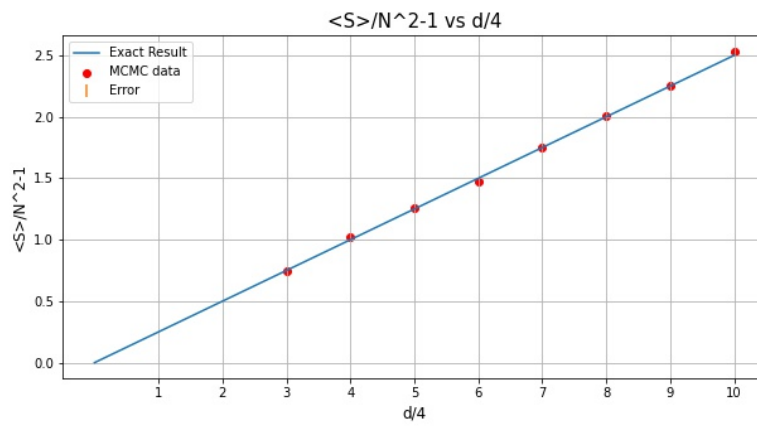


Figure 18: Plot for $N = 10$.

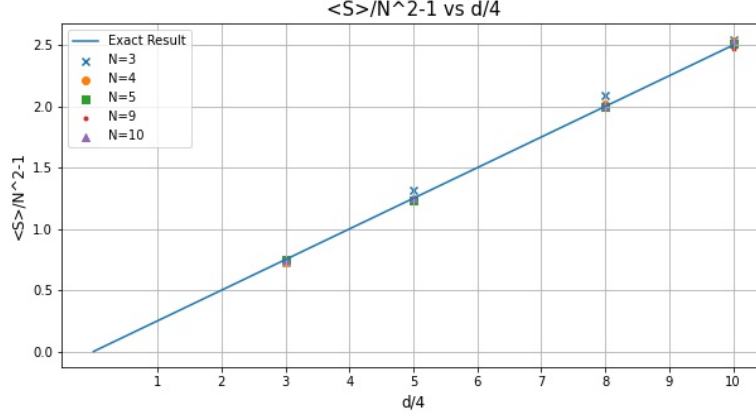


Figure 19: Combined plot for $N = 3, 4, 5, 9, 10$.

These figures show the average action for various values of N against d and the Fig. 19 shows a plot for all these value in one graph.

4 Conclusions

In this project, we have simulated the Yang-Mills Matrix Model in zero-dimensions.

Observing the plots for various values of the number of colors and number of scalars, we can see that the action generated using the Metropolis algorithm agrees very well with the analytical result.

References

- [1] B. Ydri, “Computational Physics: An Introduction to Monte Carlo Simulations of Matrix Field Theory,” doi:10.1142/10283 [arXiv:1506.02567 [hep-lat]].
- [2] A. Joseph, “Markov Chain Monte Carlo Methods in Quantum Field Theories: A Modern Primer,” SpringerBriefs in Physics, Springer (2020) doi:10.1007/978-3-030-46044-0 [arXiv:1912.10997 [hep-th]].